

Abstract

We present a domain-based local pair natural orbital (DLPNO) formulation of the algebraic diagrammatic construction scheme for the one-particle Green’s function through third order, DLPNO-IP-ADC(3), for closed-shell reference determinants. The 2h1p space is spanned by pair natural orbitals constructed over projected-atomic-orbital domains of localized occupied orbitals, so that each occupied pair carries its own compact virtual space and no quantity is ever expressed in the canonical virtual basis.

The central difficulty of a local ADC(3), which does not arise at second order, is that the third-order self-energy and the first-order 2h1p–2h1p block contain contractions in which a virtual index is summed between a two-electron integral and an amplitude belonging to a *different* occupied pair. We show that such an index must be carried in a pair space that owns it — one containing the occupied label the virtual belongs to — and that the otherwise natural choice of projecting the amplitude into the target pair introduces an error that does not vanish as the pair natural orbital threshold is tightened. Every term of the ADC(3) matrix is classified accordingly, and the routing that follows is given explicitly.

We further describe the three-step pair prescreening used to select strong pairs, the population-based criterion used to build local fitting domains, the metric that renders the spin-adapted secular problem symmetric, and a composite correction that measures the truncation error of the doubles space by folding a second-order, scaled-opposite-spin variant of the same equations both canonically and locally. Two independent implementations are described: a production code in C++ and an implementation of the same equations in dense linear algebra, retained as a check on the working equations rather than on their programming.

A domain-based local pair natural orbital formulation of third-order algebraic diagrammatic construction theory for ionization potentials

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1 Introduction

The algebraic diagrammatic construction (ADC) scheme for the one-particle Green’s function provides a systematic, size-consistent and Hermitian hierarchy of approximations to vertical ionization potentials.[1, 2, 3, 4] Its non-Dyson formulation replaces the frequency-dependent self-energy by a static Hermitian secular matrix, so that ionization energies are obtained as eigenvalues rather than as roots of a nonlinear equation, and a Davidson procedure suffices. At second order, IP-ADC(2), the method is comparable in cost to second-order Møller–Plesset theory and in accuracy to it; at third order, IP-ADC(3), it approaches the accuracy of equation-of-motion coupled cluster theory for the principal ionization peaks,[3, 4] at an $O(N^6)$ cost that confines it to small molecules.

The cost of any such method is dominated by the delocalization of the canonical molecular orbitals. Local correlation methods[5, 6] exploit the fact that the correlation energy is a sum of essentially local contributions once the occupied orbitals are localized: the correlation of a given occupied pair involves only virtual functions in its neighbourhood, and pairs of distant occupied orbitals contribute negligibly. Pair natural orbitals (PNOs) push this further by compressing the virtual space of each pair to the eigenvectors of its own first-order density,[7] and the domain-based local pair natural orbital (DLPNO) family combines the two: PNOs are constructed within projected-atomic-orbital (PAO) domains chosen by orbital overlap, giving methods whose cost grows linearly with system size while reproducing the canonical result to within a controlled and systematically improvable error.[8, 9, 10, 11]

DLPNO ideas have been carried into excited- and ionized-state theory, including the equation-of-motion coupled cluster treatment of ionization potentials,[12] where the target quantity is an eigenvalue of an effective Hamiltonian rather than an energy expectation value. The present work applies them to IP-ADC. The second-order case is close to the DLPNO-MP2 situation: the static self-energy is a contraction of first-order amplitudes with exchange integrals, and both belong to the same pair. Third order is qualitatively different. The third-order static self-energy $\mathbf{M}^{\text{ss},(3)}$ contains fourteen distinct contractions, the coupling blocks acquire third-order contributions, and the 2h1p–2h1p block acquires a first-order coupling between different pairs. Several of these terms sum a virtual index between a two-electron integral and an amplitude that belongs to a different occupied pair, and it is precisely there that a naive localization fails.

The failure is worth stating plainly, because it is silent. Every virtual index in these expressions belongs to a definite occupied label: in t_{ab}^{ij} the index a belongs to i and b to j , and the same is true of $(ia|jb)$. The PNO space of a pair spans the virtual functions of its own two occupied orbitals, and nothing else. Projecting an amplitude wholesale into a third pair — which is what a direct transcription of the canonical expressions into the local basis amounts to — therefore truncates the summed index in a space that has no reason to span it. The resulting error is not a PNO

truncation error: it does not decrease when the PNO threshold is tightened. Section 4 states the rule that avoids it, classifies every term of the ADC(3) matrix by it, and gives the resulting evaluation scheme.

The remainder of the paper is organized as follows. Section 2 collects the working equations of IP-ADC(n) for a closed-shell reference in the spin-adapted form used here. Section 3 constructs the local orbital spaces: localized occupied orbitals, projected atomic orbitals, differential-overlap domains, pair natural orbitals and the overlaps that connect them. Section 4 develops the routing rule and applies it term by term. Section 5 treats the first- and second-order amplitudes, whose local equations are coupled rather than diagonal. Section 6 describes the three-step pair prescreening and the weak-pair correction, and Section 7 the local density fitting and the criterion that sets its domains. Section 8 gives the metric and the eigensolver, and Section 9 a composite correction for the truncation of the doubles space. Section 10 describes the implementation.

2 IP-ADC for a closed-shell reference

We use i, j, k, l, m, n for occupied and a, b, c, d, e, f for virtual spatial orbitals, and $\tilde{\mu}, \tilde{\nu}$ for projected atomic orbitals. Two-electron integrals are in chemists' notation, $(ia|jb) = \int \phi_i(1)\phi_a(1)r_{12}^{-1}\phi_j(2)\phi_b(2) d1 d2$. The Fock matrix in the occupied block is F_{ij} ; it is not diagonal once the occupied orbitals are localized, and this is responsible for much of what follows.

The non-Dyson IP-ADC secular matrix is expressed in the basis of one-hole (1h) configurations $|i\rangle$ and two-hole-one-particle (2h1p) configurations $|jka\rangle$,

$$\mathbf{M} = \begin{pmatrix} \mathbf{M}^{\text{ss}} & \mathbf{M}^{\text{sd}} \\ \mathbf{M}^{\text{ds}} & \mathbf{M}^{\text{dd}} \end{pmatrix}, \quad \mathbf{M} \mathbf{X} = -\mathbf{X} \mathbf{\Omega}, \quad (1)$$

with ionization potentials $\mathbf{\Omega}$. In the spin-adapted closed-shell form the doubles configurations are not orthonormal in the sense required for a symmetric problem; the appropriate metric is

$$N_{jka, lmb} = (2\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl})\delta_{ab}, \quad (2)$$

and the two coupling blocks are related by $\mathbf{M}^{\text{sd}} = \mathbf{N} \mathbf{M}^{\text{ds}}$, so that only one of them need be constructed. We return to the consequences in Section 8.

2.1 The blocks order by order

Through first order the singles-singles block is the occupied Fock matrix,

$$M_{ij}^{\text{ss},(0)+(1)} = F_{ij}, \quad (3)$$

which in a localized basis is not diagonal. The second-order contribution is the static part of the self-energy,

$$M_{ij}^{\text{ss},(2)} = \sum_{lde} (\varepsilon_d + \varepsilon_e - \varepsilon_l) t_{de}^{il} \tilde{t}_{de}^{jl} + \left[\sum_{lde} K_{de}^{il} \tilde{t}_{de}^{jl} - \frac{1}{2} \varepsilon_i \sum_{lde} t_{de}^{il} \tilde{t}_{de}^{jl} + (i \leftrightarrow j) \right], \quad (4)$$

where $K_{ab}^{ij} = (ia|jb)$, t_{ab}^{ij} is the first-order (MP1) amplitude and $\tilde{t}_{ab}^{ij} = 2t_{ab}^{ij} - t_{ba}^{ij}$. In a localized basis the orbital energies $\varepsilon_i, \varepsilon_l$ appearing in Eq. (4) are replaced by contractions with the occupied Fock matrix,

$$\varepsilon_i t_{de}^{il} \longrightarrow \sum_k F_{ik} t_{de}^{kl}, \quad (5)$$

which couples each pair to its neighbours; $\varepsilon_d + \varepsilon_e$ remains diagonal because the virtual functions of each pair are semicanonicalized within that pair (Section 3).

The third-order contribution $M_{ij}^{\text{ss},(3)}$ consists of fourteen distinct contractions. Written compactly, and with $\hat{t}$ denoting the second-order amplitude and $t^{(1)\vee}$ the particle-ladder contraction of the first-order amplitude,

$$M_{ij}^{\text{ss},(3)} = 2 \sum_{ld} \hat{t}_{ld}^{(1)} \left[2(l\tilde{d}|ji) - (j\tilde{d}|li) \right] \quad (\text{T1})$$

$$+ 2 \sum_{lde} \tilde{t}_{de}^{il} (j\tilde{d}|l\tilde{e}) \quad (\text{T2})$$

$$+ 2 \sum_{lde} t_{de}^{il} (\varepsilon_d + \varepsilon_e - \varepsilon_j - \varepsilon_l) \tilde{t}_{de}^{jl} \quad (\text{T3})$$

$$+ 2 \sum_{lm} \sum_{def} \tilde{t}_{de}^{lm} \tilde{t}_{df}^{lj} (m\tilde{e}|i\tilde{f}) \quad (\text{T4})$$

$$+ \sum_{lm} \sum_{def} \tilde{t}_{de}^{il} (l\tilde{e}|m\tilde{f}) \tilde{t}_{df}^{jm} \quad (\text{T5})$$

$$- 2 \sum_{ef} \Theta_{ef}^{mj} (mi|f\tilde{e}) \quad (\text{T6})$$

$$- \sum_{mef} t_{fe}^{im} \left[\sum_{ld} \tilde{t}_{fd}^{jl} (ml|\tilde{d}\tilde{e}) \right] \quad (\text{T7})$$

$$- \sum_{mef} t_{fe}^{mi} \left[\sum_{ld} \tilde{t}_{fd}^{lj} (ml|\tilde{d}\tilde{e}) \right] \quad (\text{T8})$$

$$+ 2 \sum_{fe} G_{fe}^{lm} (ji|\tilde{e}\tilde{f}) - \sum_{fe} G_{fe}^{lm} (jf|i\tilde{e}) \quad (\text{T9,T10})$$

$$- 2 \sum_{mn} \Lambda_{mn} (ji|nm) + \sum_{mn} \Lambda_{mn} (jm|ni) \quad (\text{T11,T12})$$

$$+ 2 \sum_{lnm} \left[\sum_{de} t_{de}^{in} \tilde{t}_{de}^{lm} \right] (jl|nm) \quad (\text{T13})$$

$$+ \sum_{lde} \tilde{t}_{de}^{il} t_{de}^{(1)\vee, jl}, \quad (\text{T14})$$

with the intermediates

$$\Theta_{ef}^{mj} = \sum_{ld} \left[t_{de}^{lm} \tilde{t}_{df}^{lj} + t_{ed}^{lm} \tilde{t}_{df}^{jl} \right], \quad (6)$$

$$G_{fe}^{lm} = \sum_d t_{df}^{lm} \tilde{t}_{de}^{lm}, \quad (7)$$

$$\Lambda_{mn} = \sum_{lde} t_{de}^{ln} \tilde{t}_{de}^{lm}, \quad (8)$$

$$t_{cd}^{(1)v,ij} = \sum_{ab} (ca|db) t_{ab}^{ij}. \quad (9)$$

The occupied indices l, m, n are summed and, together with i and j , label different pairs; it is this structure that Section 4 has to deal with.

The coupling blocks are, through first order,

$$M_{jka,i}^{\text{ds},(1)} = (j\tilde{a}|ki), \quad M_{i,jka}^{\text{sd},(1)} = 2(j\tilde{a}|ki) - (k\tilde{a}|ji), \quad (10)$$

and acquire at third order

$$M_{jka,i}^{\text{ds},(3)} = \sum_{bc} (i\tilde{c}|\tilde{a}\tilde{b}) t_{cb}^{kj} \quad (D)$$

$$+ \sum_{lb} \tilde{t}_{ba}^{lj} (\tilde{l}\tilde{b}|ik) \quad (A)$$

$$- \sum_{lb} t_{ba}^{lj} (i\tilde{b}|lk) \quad (B)$$

$$- \sum_{lb} t_{ba}^{kl} (i\tilde{b}|lj). \quad (C)$$

The doubles–doubles block is diagonal at zeroth order,

$$M_{jka,lmb}^{\text{dd},(0)} = (\varepsilon_j + \varepsilon_k - \varepsilon_a) \delta_{jl} \delta_{km} \delta_{ab}, \quad (11)$$

— again with $\varepsilon_j + \varepsilon_k$ replaced by contractions with F in a localized basis — and acquires at first order, which is what distinguishes ADC(3) from ADC(2) in this block,

$$\begin{aligned} \left(\mathbf{M}^{\text{dd},(1)} \mathbf{r} \right)_{jka} = & -\frac{1}{2} \sum_{il} (ki|jl) r_{lia} - \frac{1}{2} \sum_{il} (kl|ji) r_{ila} \\ & - \sum_{lb} (j\tilde{a}|\tilde{l}\tilde{b}) [2r_{lkb} - r_{klb}] \\ & + \sum_{lb} (jl|\tilde{b}\tilde{a}) r_{lkb} + \sum_{lb} (kl|\tilde{b}\tilde{a}) r_{jlb}. \end{aligned} \quad (12)$$

The two contributions on the first line are the same contraction written twice over complementary index ranges and are evaluated together.

3 Local orbital spaces

3.1 Occupied and virtual bases

The occupied orbitals are localized by the Pipek–Mezey criterion.[13] Frozen core orbitals are excluded from the localization and from the correlation treatment. The virtual space is spanned

by projected atomic orbitals,

$$|\tilde{\mu}\rangle = \left(1 - \sum_i^{\text{all occ}} |i\rangle \langle i|\right) |\mu\rangle, \quad (13)$$

where the projector runs over *all* occupied orbitals including the frozen core; projecting only the correlated occupied space would leave core density in the virtual space and inflate every domain. PAOs are non-orthogonal and linearly dependent, with overlap $S^{\tilde{\mu}\tilde{\nu}}$.

3.2 Domains

Locality is measured by the differential overlap integral[10]

$$\text{DOI}(i, \tilde{\mu}) = \left[\int |\phi_i(\mathbf{r})|^2 |\phi_{\tilde{\mu}}(\mathbf{r})|^2 d\mathbf{r} \right]^{1/2}, \quad (14)$$

evaluated on a numerical grid. An atom A belongs to the domain of i if any PAO centred on A satisfies $\text{DOI}(i, \tilde{\mu}) > \text{TCUTDO}$, and the domain $[i]$ consists of all PAOs on the atoms so selected. Domains are atom-granular by construction: a domain that cut through an atom's basis set would not be invariant to rotations of that basis. The domain of a pair is the union $[ij] = [i] \cup [j]$.

Within a pair domain the PAOs are made orthonormal and semicanonical in two steps. The domain overlap is diagonalized and eigenvectors with eigenvalues below TCUTPAO are discarded, removing the linear dependence; the Fock matrix is then diagonalized in the surviving space. We call the result the *non-redundant semicanonical domain basis* of the pair, of dimension n_{nr}^{ij} , and denote its transformation matrix $\mathbf{d}_{\text{nr}}^{ij}$. It is the largest space the pair can represent, and it will be needed in Section 5.

3.3 Pair natural orbitals

In that basis the semicanonical first-order amplitude is

$$\bar{t}_{ab}^{ij} = \frac{-K_{ab}^{ij}}{\varepsilon_a + \varepsilon_b - F_{ii} - F_{jj}}, \quad (15)$$

and the pair density is

$$\mathbf{D}^{ij} = \frac{2}{1 + \delta_{ij}} \left[\tilde{\mathbf{t}}^\dagger \tilde{\mathbf{t}} + \tilde{\mathbf{t}} \tilde{\mathbf{t}}^\dagger \right]. \quad (16)$$

Its eigenvectors with occupation numbers above TCUTPNO are retained as the pair natural orbitals of (i, j) ; the Fock matrix is diagonalized once more within the retained space so that ε_a remains a genuine orbital energy there. Writing $\mathbf{v}^{ij}$ for the retained eigenvectors, the PNO coefficients in the domain are

$$\mathbf{d}^{ij} = \mathbf{d}_{\text{nr}}^{ij} \mathbf{v}^{ij}, \quad (17)$$

a factorization we use repeatedly: $\mathbf{d}_{\text{nr}}^{ij}$ spans the whole domain, $\mathbf{d}^{ij}$ only the part of it the pair correlates into. The unordered pair $\{i, j\}$ carries one PNO space, shared by the two ordered labels (i, j) and (j, i) , since the density (16) is symmetric under the exchange even though the amplitude is not.

3.4 Cross-pair overlaps

Quantities belonging to different pairs are related by the overlap of their PNO spaces,

$$S_{ab}^{PQ} = \langle a_P | b_Q \rangle = [(\mathbf{d}^P)^\dagger \mathbf{S}^{\text{PAO}} \mathbf{d}^Q]_{ab}, \quad (18)$$

where $\mathbf{S}^{\text{PAO}}$ is restricted to the union of the two domains. A matrix $\mathbf{X}$ living in pair Q is expressed in pair P as $\mathbf{S}^{PQ} \mathbf{X} (\mathbf{S}^{PQ})^\dagger$. This transformation is exact if and only if $\mathbf{X}$ has no support outside the space $\mathbf{d}^P$ spans — a condition that is the subject of the next section.

4 Evaluating the ADC matrix in the local basis

4.1 The rule

Consider a contraction of the generic form

$$X = \sum_c g_{\dots c \dots} A_{\dots c \dots}^Q, \quad (19)$$

in which a virtual index c is summed between a two-electron integral g and an amplitude A belonging to pair Q , and the result X is required in the space of some other pair P . Two evaluations suggest themselves. One may transform A into P first and contract there, which requires the integral only in P 's space; or one may contract in Q , where A is exact, and transform the result.

The two are not equivalent, and the choice is not a matter of taste. Every virtual index in Section 2 is associated with a definite occupied label — a with i and b with j in both t_{ab}^{ij} and $(ia|jb)$ — and the PNO space of a pair is constructed, through Eqs. (15)–(16), to span the virtual functions its own two occupied orbitals correlate into. It has no claim on the virtual functions of a third orbital. We therefore adopt the following rule.

Rule. *A virtual index summed between a two-electron integral and an amplitude must be carried in a pair space that owns it, i.e. one whose occupied labels include the occupied label the index belongs to.*

Two classes of contraction are exempt, and it is useful to name them.

(i) *Amplitude–amplitude contractions.* If the summed index connects two amplitudes with no integral between them, both factors decay with the index, and the projection $\mathbf{S}^{PQ} \mathbf{X} (\mathbf{S}^{PQ})^\dagger$ into either pair is accurate to the PNO truncation of the amplitudes themselves. Terms T11–T13 and the Λ_{mn} , G^{lm} intermediates are of this kind.

(ii) *Coulomb-type integrals.* If the two virtual indices of an integral sit on the same electron, as in $(xy|\tilde{a}\tilde{b})$, the integral does not assign them to separate occupied labels and both must in any case share a space. The choice is then determined by the surrounding amplitudes rather than by the integral.

Everything else falls under the rule, and violating it has a characteristic signature: because the neglected part of the summand is not the tail of a converging PNO expansion but a piece of a different pair's space, the error does *not* decrease as TCUTPNO is tightened. This is what makes the mistake easy to miss — the calculation looks converged.

4.2 Term-by-term classification

We now apply the rule to each block. Throughout, P denotes the target pair and $\rightarrow$ a transformation by $\mathbf{S}^{PQ}$.

Terms native to one pair. T2 contracts $\tilde{t}^{il}$ with $(jd|\tilde{l}\tilde{e})$, which is the *ovov* block of the pair (i, l) itself; T9 and T10 contract two amplitudes of the same pair (l, m) with integrals of that pair; the particle ladder in $t^{(1)\vee}$ never leaves the pair it belongs to. These are evaluated in place, with no transformation at all.

Terms of class (i). T3 contracts t^{il} with a function of $\tilde{t}^{jl}$; both carry the index l , and the transformation $(j, l) \rightarrow (i, l)$ is exact in the sense of (i). T11–T13 likewise. The occupied-Fock substitutions of Eq. (5) are of the same character.

T4 and T6: intermediates whose indices live in different pairs. In T4 the index d is summed between $\tilde{t}^{lm}$ and $\tilde{t}^{lj}$ and belongs to l ; e belongs to m and f to j , while the integral $(m\tilde{e}|i\tilde{f})$ belongs to the pair (m, i) . Building the intermediate in (m, j) — the choice suggested by its occupied labels, and the one that would allow the l sum to be accumulated once and reused for every i — places both d and f in a pair that owns neither. Instead we form

$$\mathcal{G}_{ef}^{(l,m),(l,j)} = \sum_d \tilde{t}_{de}^{lm} \left[\mathbf{S}^{(lm)(lj)} \tilde{\mathbf{t}}^{lj} \right]_{df}, \quad (20)$$

whose two indices lie in *different* pairs, each of which owns its index, and only then map it into the integral's pair, $\mathcal{G} \rightarrow \mathbf{S}^{(mi)(lm)} \mathcal{G} (\mathbf{S}^{(mi)(lj)})^\dagger$. The price is that the l summation can no longer be pre-accumulated and the loop is one factor of n_{occ} deeper; the benefit is that the term converges. T6 is treated identically, with the second half of Θ^{mj} carrying e in (l, m) and f in (j, l) .

T5: a contested index contracted directly. Here e belongs to l and f to m , so both may be brought to the integral's pair (l, m) , which owns them. The index d , however, belongs to i in one amplitude and to j in the other, and (l, m) owns neither. It is therefore never represented there: with

$$\mathcal{A}_{df}^i = \left[\tilde{\mathbf{t}}^{il} \mathbf{S}^{(il)(lm)} \mathbf{K}^{lm} \right]_{df}, \quad (21)$$

$$\mathcal{B}_{df}^j = \left[\tilde{\mathbf{t}}^{jm} \mathbf{S}^{(jm)(lm)} \right]_{df}, \quad (22)$$

in which only the index the integral owns has been mapped, the term is

$$\text{T5} = \sum_{lm} \left\langle \mathcal{A}^i (\mathcal{B}^j)^\dagger, \mathbf{S}^{(il)(jm)} \right\rangle, \quad (23)$$

the contested index being contracted directly between the two amplitudes' own spaces.

T7 and T8: choosing the integral's home. The integral $(ml|\tilde{d}\tilde{e})$ is of class (ii), but the amplitudes around it still tie d to l and e to m . The pair that owns both is (l, m) , and that is where it is read: it is that pair's own *oovv* block, indexed by its own two occupied labels, so it always exists and requires no screening of occupied lists. From there d moves to (j, l) to meet $\tilde{t}^{jl}$ and e to (i, m) to meet t^{im} , each sharing an occupied label with its destination. The remaining index f is genuinely contested — it belongs to j in the amplitude and to i in the target — and is mapped once; that mapping is unavoidable and is what makes the term small when i and j are far apart.

$\mathbf{M}^{\text{ds},(3)}$. Term D keeps every index in the target pair (j, k) , which owns them all. In terms A, B and C the summed index b belongs to l (A, B) or to k (C), and the target has no claim on it: the contraction over b is therefore performed in the *amplitude's* pair, reading the *oooo* integral out of that pair's block, and only the surviving index a — which both pairs own, since they share an occupied label — is carried across with $\mathbf{S}^{PQ}$.

The first-order doubles block. In Eq. (12) the two *oooo* contractions involve no integral over virtual indices, so transformation into the target pair is exact in the sense of (i). The remaining three do: in $(j\tilde{a}|l\tilde{b})$ the index b belongs to l and is summed against r_{lkb} , and the target (j, k) has no claim on it. Each is therefore evaluated in the pair that owns b — (j, l) for the exchange and the first Coulomb line, (k, l) for the second — and mapped back afterwards. Two transformations replace one, but both are between pairs sharing an occupied index with the index being moved.

Screening. The routing determines the screening rather than the other way round. In T4, for example, l contributes only if (l, m) and (l, j) are both strong pairs, and i only if (m, i) is one; no pair (i, j) is ever required, so M_{ij}^{ss} survives for i, j that do not themselves form a strong pair, as it must.

5 Amplitudes in the local basis

5.1 First-order amplitudes

Because F_{ij} is not diagonal, the first-order amplitudes satisfy a coupled system rather than a division,

$$R_{ab}^{ij} = K_{ab}^{ij} - (F_{ii} + F_{jj} - \varepsilon_a - \varepsilon_b) t_{ab}^{ij} - \sum_{k \neq i} F_{ik} [\mathbf{S} \mathbf{t}^{kj} \mathbf{S}^\dagger]_{ab} - \sum_{k \neq j} F_{jk} [\mathbf{S} \mathbf{t}^{ik} \mathbf{S}^\dagger]_{ab}, \quad (24)$$

solved iteratively with the semicanonical denominator as preconditioner. The coupling is of class (i) and the transformations are exact.

5.2 Second-order amplitudes

The second-order amplitude obeys the same coupled system with a different right-hand side,

$$\begin{aligned} \mathcal{D}_{ab}^{ij} = & \sum_{cd} (ca|db) t_{cd}^{ij} + \sum_{kl} (ki|lj) t_{ab}^{kl} \\ & + 2 \sum_{kc} (kc|bj) t_{ca}^{ki} - \sum_{kc} (kc|bj) t_{ca}^{ik} \\ & - \sum_{kc} (kj|bc) t_{ac}^{ik} - \sum_{kc} (ki|bc) t_{ac}^{kj} - \sum_{kc} (kj|ac) t_{cb}^{ik} \\ & + 2 \sum_{kc} (kc|ai) t_{cb}^{kj} - \sum_{kc} (kc|ai) t_{cb}^{jk} - \sum_{kc} (ki|ac) t_{cb}^{kj}. \end{aligned} \quad (25)$$

Solving Eq. (25) by division, as the canonical expression permits, discards the inter-pair Fock coupling and is a measurable error; the coupled system is solved instead.

The routing question here is different from that of Section 4, because the target indices a, b are being *constructed* rather than met: there is no third pair that owns the summed index c . The first

two terms are safe — the particle ladder never leaves the pair, and the *oooo* term sums no virtual against an integral — and are evaluated in the pair’s PNO space. In the remaining eight, c belongs to the summed occupied label k and the amplitude comes from a neighbouring pair. These are assembled in the pair’s non-redundant semicanonical domain basis $\mathbf{d}_{\text{nr}}^{ij}$ of Eq. (17), which spans the whole domain and therefore truncates nothing the neighbour brings, and projected into the PNO space only once, at the end:

$$\mathcal{D}^{ij} += (\mathbf{v}^{ij})^\dagger \mathcal{D}_{\text{nr}}^{ij} \mathbf{v}^{ij}. \quad (26)$$

5.3 Second-order singles

Term T1 of $M^{\text{ss},(3)}$ requires the second-order singles amplitude, whose driver is

$$\mathcal{D}_{ia}^{(1)} = \sum_{kcd} \tilde{t}_{cd}^{ik} (k\tilde{d}|\tilde{a}\tilde{c}) - \sum_{ick} (i\tilde{c}|kj) \tilde{t}_{cd}^{ik} \Big|_{jd}, \quad (27)$$

and which is represented in the PNO space of the diagonal pair (i, i) . As with Eq. (24) the equation couples different i through F and is solved as a linear system.

6 Pair prescreening and the weak-pair correction

A pair that is classified strong carries a PNO space, an amplitude, four integral blocks and a slice of the ADC eigenvector; a pair that is not costs nothing. The classification is made in three steps.

Step 1: a cheap estimate over every pair. Two channels are used, because no single inexpensive quantity sees both sources of a pair energy.

The *overlap channel* is the one that decides on ordinary molecules. When two localized orbitals share space the pair energy is driven by their exchange integral, which involves the product density $\phi_i(\mathbf{r})\phi_j(\mathbf{r})$; the natural scaling is therefore quadratic in the differential overlap,

$$e_{ij} \approx -c_{\text{DOI}} \text{DOI}(i, j)^2. \quad (28)$$

Measured over every occupied pair of a four-water chain at 3.2 Å and of a C₈H₁₆ chain, both in cc-pVDZ, the ratio $|e_{ij}|/\text{DOI}(i, j)^2$ lies between 0.3 and 10^2 across all pairs worth more than $10^{-6} E_h$ — two orders of magnitude of scatter for pair energies spanning five. The smallest prefactor that would retain every pair worth more than TCUTPAIRS, at TCUTPRE = TCUTPAIRS/100, is 0.38 on those systems; we use $c_{\text{DOI}} = 25$, a safety factor of roughly sixty, and the resulting estimate of the discarded correlation energy agrees with the true value to within about ten percent.

The *dispersion channel* covers what Eq. (28) cannot see. Two orbitals that do not overlap at all still have an R^{-6} pair energy, and the differential overlap is exponentially blind to it. There the exchange integral is dominated by the dipole–dipole interaction of the two transition densities, giving the semicanonical estimate

$$e_{ij}^{\text{dip}} = -\frac{4}{R_{ij}^6} \sum_{a \in [i]} \sum_{b \in [j]} \frac{[\boldsymbol{\mu}_{ia} \cdot \boldsymbol{\mu}_{jb} - 3(\boldsymbol{\mu}_{ia} \cdot \hat{\mathbf{R}})(\boldsymbol{\mu}_{jb} \cdot \hat{\mathbf{R}})]^2}{\varepsilon_a + \varepsilon_b - F_{ii} - F_{jj}}, \quad (29)$$

with $\boldsymbol{\mu}_{ia} = \langle i | \mathbf{r} | \tilde{a} \rangle$ and $\mathbf{R}_{ij}$ the vector between the centroids of i and j ; a and b run over each orbital’s own domain, orthogonalized and semicanonicalized, so that ε_a is a genuine orbital energy of that domain. Both the transition dipoles and the centroid difference are origin-independent, the former

because the PAO projector (13) makes PAOs orthogonal to all occupied orbitals. Equation (29) is formed only for pairs whose differential overlap lies below TCutDO_{ij} , since a multipole expansion between overlapping distributions is meaningless.

A pair is retained if *either* channel exceeds TCutPRE in magnitude; neither is a veto. In practice the overlap channel does nearly all of the work — on the systems above the dispersion channel alters one decision in several hundred — but it is inexpensive and the situation it covers, a pair of well-separated polarizable fragments, is a real one.

The estimate is evaluated for every unordered pair rather than for a pre-filtered subset. It is $O(1)$ per pair, so there is nothing to be gained by filtering ahead of it, and a topological filter — retaining only pairs whose domains share an atom, say — would discard pairs without an energy attached to them, leaving a hole in the accounting of Eq. (31) below.

Step 2: semicanonical pair energies. For every survivor the semicanonical local MP2 pair energy is computed in the pair’s own domain,

$$e_{ij} = (2 - \delta_{ij}) \sum_{ab} K_{ab}^{ij} \left(2\bar{t}_{ab}^{ij} - \bar{t}_{ba}^{ij} \right), \quad (30)$$

with $\bar{t}$ from Eq. (15). This is the quantity TCutPAIRS is applied to. It costs one domain orthogonalization, one local fit and one division per candidate — the same work as the first steps of the PNO construction, without the PNOs — and it is a far better criterion than Eq. (29), which is blind to overlap-driven pair energies by construction.

Step 3: the weak-pair correction. The pair energies of the discarded pairs are summed,

$$E_{\text{weak}} = \sum_{ij \in \text{weak}} e_{ij} + \sum_{ij \in \text{step 1}} e_{ij}^{\text{est}}, \quad (31)$$

using the best estimate available for each: the semicanonical pair energy for pairs step 2 rejected, and whichever of the two step-1 channels was larger in magnitude for those rejected earlier. E_{weak} is a real approximation to the correlation energy discarded and is reported alongside the local MP2 energy; their sum estimates the unscreened result.

7 Local density fitting

All two-electron integrals are obtained from a density-fitting approximation,[15, 16]

$$(pq|rs) \approx \sum_{KL} (pq|K) [\mathbf{J}^{-1}]_{KL} (L|rs), \quad (32)$$

in which the auxiliary index is restricted to a fitting domain that depends on the pair. The functions a pair’s fit must represent are the products $\phi_i(\mathbf{r})\phi_{\bar{\mu}}(\mathbf{r})$, whose extent is governed by the *occupied* orbital; the PAO domain answers a different question and is not the right criterion. We therefore select fitting atoms from the orbital’s own atomic populations, in the spirit of Mata and Werner:[14] for each localized occupied orbital the atoms are ordered by decreasing Löwdin population of $|\phi_i|^2$ and taken in that order until the population left behind falls below TCutMKN ; the fitting domain of a pair is the union of its two orbitals’ atom sets, and the auxiliary functions on them. A Löwdin rather than a Mulliken partition is used because it is positive, so that “the population left behind”

is a quantity a threshold can meaningfully be applied to. Smaller TCUTMKN retains more atoms, in the same direction as every other threshold in the method.

A local fitting domain is a subset of an auxiliary basis that was never designed to be complete on its own, so the restricted Coulomb metric is routinely ill-conditioned. Its inverse is therefore formed as a pseudo-inverse from the eigendecomposition, with directions below a relative cutoff discarded; a Cholesky or LU factorization does not fail on such a matrix but returns a fit that is wrong in the tails.

8 The metric and the eigensolver

The metric (2) is block diagonal over unordered pairs in the PNO basis, each block acting on the pair of ordered labels (j, k) and (k, j) as $2\mathbf{1} - \mathbf{P}$ with $\mathbf{P}$ the exchange of the two. Its eigenvalues are therefore 1 on the symmetric combination and 3 on the antisymmetric one, and $\mathbf{N}^{\pm 1/2}$ is available in closed form. The similarity transformation

$$\mathbf{N}^{-1/2} \mathbf{M} \mathbf{N}^{-1/2} \mathbf{y} = -\Omega \mathbf{y}, \quad \mathbf{y} = \mathbf{N}^{1/2} \mathbf{x}, \quad (33)$$

renders the problem symmetric at no cost, and a symmetric Davidson procedure[17] is used, with the diagonal of $\mathbf{M}^{\text{dd}}$ as preconditioner — which remains diagonal precisely because the PNOs of each pair are semicanonicalized.

One caveat is worth recording. The third-order static self-energy as conventionally written is not symmetric, although the exact self-energy is Hermitian; the asymmetry is an omission in the working equations rather than physics. It is small — it moves the roots by well under 1 meV — but it breaks the self-adjointness the symmetric Davidson relies on, so that the residual cannot fall below the asymmetry and the iteration stalls short of its tolerance. $\mathbf{M}^{\text{ss},(3)}$ is therefore symmetrized before diagonalization.

9 A composite correction for the doubles space

The dominant approximation of the method is the truncation of the 2h1p space to the PNOs of each pair. At strict second order this error can be measured rather than estimated. The 2h1p–2h1p block of IP-ADC(2) is Eq. (11) alone, so Löwdin partitioning of Eq. (1) is an identity, not an approximation:[18]

$$[\mathbf{M}^{\text{ss}} + \mathbf{F}(\omega)] \mathbf{c} = \omega \mathbf{c}, \quad \mathbf{F}(\omega) = \mathbf{M}^{\text{sd}} \left(\omega - \mathbf{M}^{\text{dd}} \right)^{-1} \mathbf{M}^{\text{ds}}, \quad (34)$$

with $\Omega = -\omega$ and no doubles vector ever formed. Canonically $(\omega - \mathbf{M}^{\text{dd}})^{-1}$ is a scalar division; in the PNO basis only the ε_a part is diagonal and the two occupied indices become Fock couplings to neighbouring pairs, so Eq. (34) becomes a shifted linear system in the doubles space, which is solved rather than approximated — dropping the coupling would contaminate the difference we are trying to measure.

Running the fold canonically, before localization, and again in the PNO basis gives the second-order truncation error directly. We use the scaled-opposite-spin variant of the second-order self-energy for this purpose, in which the eight terms of Eq. (4) collapse to two scaled by c_{os} , the coupling blocks and $\mathbf{M}^{\text{dd}}$ being untouched. The difference is added to the third-order result as a composite correction. It removes the doubles-space channel exactly; it cannot touch the third-order amplitude channel, and whether the composite is an improvement is therefore an empirical question.

10 Implementation

The method is implemented in the BAGH program package. Orbital localization, the projected atomic orbitals, the differential-overlap and dipole matrices and the three-index integrals are prepared in Python on top of PySCF;[19] everything after that — domains, prescreening, pair natural orbitals, the integral store, the amplitude equations, the intermediates and the Davidson procedure — is carried out in C++ behind a pybind11 interface, parallelized with OpenMP. No trial vector, sigma vector or amplitude crosses the language boundary.

Sparsity is represented throughout by sparse maps in the sense of Pinski *et al.*[10] All two-electron integrals are generated once, after the PNO spaces are complete, directly in the PNO basis of the pair that owns them; nothing is generated on the fly and no integral is ever stored in a canonical or PAO basis. For each unordered pair U the store holds $(x\tilde{a}|yb)$, $(xy|\tilde{a}\tilde{b})$, $(x\tilde{a}|yz)$ and $(x\tilde{a}|\tilde{b}\tilde{c})$, the occupied labels x, y, z running over a list comprising the pair’s own two labels together with those whose differential overlap with the pair’s domain exceeds a threshold.

Because the largest of these classes can exceed available memory while the smallest cannot, residency is planned rather than left to the operating system. Before any integral is formed, the size of each class is computed from the PNO dimensions and compared against the user’s memory limit; classes are assigned to memory or to a memory-mapped file in order of access frequency, with the pair data, the Davidson subspace and the *oooo* block never spilled. A calculation whose unspillable data exceeds the limit stops with a report rather than paging.

A second, independent implementation of the same equations is maintained alongside the first. It is written in dense linear algebra with no sparse maps, no memory management and no Davidson procedure: every quantity is a dense array, each contraction is a single tensor contraction, and the full ADC matrix is formed and diagonalized directly. It shares no code with the production implementation, and reproduces it to 10^{-3} meV in the ionization potentials and $10^{-12} E_h$ in the second- and third-order energies at every threshold tested. Its purpose is to check the working equations rather than their programming, and it was in fact the means by which two of the routing choices of Section 4 were found to be suboptimal in the production code.

Author declarations

The authors have no conflicts to disclose.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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